# Scale Exercises - Section II, Level 19

## Major Scale Diatonic Triads: Two-Octave Six-String Neck Ladder

*22 triads. 66 notes. One harmonic line.*

Build the scale into harmony: stack every other degree, hear each chord quality, then shift one fret.

### 1. Stack every other tone

Play 1-3-5, then begin the next triad on 2. Each group is a broken diatonic chord, not a new scale shape.

### 2. Hear the chord family

The major scale produces major, minor, and diminished triads in the fixed order I-ii-iii-IV-V-vi-vii-dim.

### 3. Shift only after note 66

Finish the descending 5-3-1 triad, release pressure, and move the complete hand-and-thumb frame exactly one fret.

## Worked example - A major at low-E fret 5

Scale: A B C# D E F# G# A B C# D E F# G# A

Diatonic triads: A | Bm | C#m | D | E | F#m | G#dim

**UP: 1-3-5 | 2-4-6 | ... | 11-13-15**

**DOWN: 15-13-11 | 14-12-10 | ... | 5-3-1**

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I-------|ii------|iii-----|IV------|V-------|vi------|vii-dim-|I-------|ii------|iii-----|IV------|
Deg|1--3--5-|2--4--6-|3--5--7-|4--6--8-|5--7--9-|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4-|4--2--1-|1--4--3-|2--1--4-|4--3--1-|1--4--3-|3--1--4-|4--3--2-|1--4--4-|3--2--1-|4--4--2-|
e  |--------|--------|--------|--------|--------|--------|--------|--------|--------|------4-|------5-|
B  |--------|--------|--------|--------|--------|--------|--------|------5-|------7-|---5-----|---7-----|
G  |--------|--------|--------|--------|------4-|------6-|---4--7-|---6-----|4--7-----|6--------|7--------|
D  |--------|------4-|------6-|---4--7-|---6-----|4--7-----|6--------|7--------|--------|--------|--------|
A  |---4--7-|---5-----|4--7-----|5--------|7--------|--------|--------|--------|--------|--------|--------|
E  |5--------|7--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV------|iii-----|ii------|I-------|vii-dim-|vi------|V-------|IV------|iii-----|ii------|I-------|
Deg|15-13-11-|14-12-10-|13-11-9-|12-10-8-|11-9--7-|10-8--6-|9--7--5-|8--6--4-|7--5--3-|6--4--2-|5--3--1-|
Fng|2--4--4-|1--2--3-|4--4--1-|2--3--4-|4--1--3-|3--4--1-|1--3--4-|4--1--2-|3--4--1-|1--2--4-|4--1--2-|
e  |5--------|4--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
B  |---7-----|---5-----|7--------|5--------|--------|--------|--------|--------|--------|--------|--------|
G  |------7--|------6--|---7--4-|---6-----|7--4-----|6--------|4--------|--------|--------|--------|--------|
D  |---------|--------|--------|------7--|------6--|---7--4--|---6-----|7--4-----|6--------|4--------|--------|
A  |---------|--------|--------|--------|--------|--------|------7--|------5--|---7--4--|---5-----|7--4-----|
E  |---------|--------|--------|--------|--------|--------|--------|--------|--------|------7--|------5--|
```

**KEY ROUTE** A -> Bb -> B -> C -> Db -> D -> Eb -> E -> F -> Gb -> G -> Ab -> A, then reverse

**AUDIT** Hear the triad quality, preserve the triplet pulse, articulate the repeated high root, and shift silently.

## Session protocol

1. A major: first-octave triads only.
2. Complete all 11 ascending triads.
3. Complete all 11 descending triads.
4. Link the high-root turn without resetting.
5. Use 2-3 neighboring rungs; pause only after each cell.
6. Connect the full route with both pick starts.

# Level 19 - Chromatic Major-Key Route and Diatonic-Triad Map

**Ascending route: complete one 66-note triad cell, then move the frame +1 fret**

Descending route: use the same cards in reverse order and move -1 fret

| Rung | Key | Low-E root | Frame | Diatonic triads (degrees 1-7) |
|---|---|---|---|---|
| 01 | A major | fret 5 | 4-7 | A Bm C#m D E F#m G#dim |
| 02 | Bb major | fret 6 | 5-8 | Bb Cm Dm Eb F Gm Adim |
| 03 | B major | fret 7 | 6-9 | B C#m D#m E F# G#m A#dim |
| 04 | C major | fret 8 | 7-10 | C Dm Em F G Am Bdim |
| 05 | Db major | fret 9 | 8-11 | Db Ebm Fm Gb Ab Bbm Cdim |
| 06 | D major | fret 10 | 9-12 | D Em F#m G A Bm C#dim |
| 07 | Eb major | fret 11 | 10-13 | Eb Fm Gm Ab Bb Cm Ddim |
| 08 | E major | fret 12 | 11-14 | E F#m G#m A B C#m D#dim |
| 09 | F major | fret 13 | 12-15 | F Gm Am Bb C Dm Edim |
| 10 | Gb major | fret 14 | 13-16 | Gb Abm Bbm Cb Db Ebm Fdim |
| 11 | G major | fret 15 | 14-17 | G Am Bm C D Em F#dim |
| 12 | Ab major | fret 16 | 15-18 | Ab Bbm Cm Db Eb Fm Gdim |
| 13 | A major - octave | fret 17 | 16-19 | A Bm C#m D E F#m G#dim |

## How to use the map

**Up pass:**

rungs 01-13.

**Turn:**

complete the high-A cell at fret 17, then reverse.

**Down pass:**

rungs 12-01. Do not replay rung 13 unless beginning a new round.

**Learning pulse:**

one triad per beat as triplets.

**Performance pulse:**

two triads per beat as sextuplets.

**Goal:**

hear one chord family moving through every key, not isolated fret shapes.

## Sequence anatomy

```
UP:    01-03-05 | 02-04-06 | 03-05-07 | 04-06-08 | 05-07-09 | 06-08-10 | 07-09-11 | 08-10-12 | 09-11-13 | 10-12-14 | 11-13-15

DOWN:  15-13-11 | 14-12-10 | 13-11-09 | 12-10-08 | 11-09-07 | 10-08-06 | 09-07-05 | 08-06-04 | 07-05-03 | 06-04-02 | 05-03-01
```

**HARMONY** `I | ii | iii | IV | V | vi | vii-dim | I | ii | iii | IV`

**PICKING** `D-U-D | U-D-U | D-U-D | U-D-U ... ; the starting stroke flips at every triad`

## Mastery gate

Name every key, preserve the three-note pulse, hear major/minor/diminished triads, shift silently, and complete both pick starts on separate days.

# Level 19 - Major Scale Diatonic Triads: Two Octaves

**Rungs 01-02: play each card as one 22-triad cell, then move +1 fret**

For the descending pass, read the cards in reverse order and move -1 fret

## 01 A MAJOR

**ROOT E5 | FRAME 4-7**

**66 NOTES / 22 TRIADS**

Scale: A B C# D E F# G# A B C# D E F# G# A

Triads: A | Bm | C#m | D | E | F#m | G#dim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------4--|------5--|
B  |---------|---------|---------|---------|---------|---------|---------|------5--|------7--|---5-----|---7-----|
G  |---------|---------|---------|---------|------4--|------6--|---4--7--|---6-----|4--7-----|6--------|7--------|
D  |---------|------4--|------6--|---4--7--|---6-----|4--7-----|6--------|7--------|---------|---------|---------|
A  |---4--7--|---5-----|4--7-----|5--------|7--------|---------|---------|---------|---------|---------|---------|
E  |5--------|7--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |5--------|4--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---7-----|---5-----|7--------|5--------|---------|---------|---------|---------|---------|---------|---------|
G  |------7--|------6--|---7--4--|---6-----|7--4-----|6--------|4--------|---------|---------|---------|---------|
D  |---------|---------|---------|------7--|------6--|---7--4--|---6-----|7--4-----|6--------|4--------|---------|
A  |---------|---------|---------|---------|---------|---------|------7--|------5--|---7--4--|---5-----|7--4-----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------7--|------5--|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

## 02 Bb MAJOR

**ROOT E6 | FRAME 5-8**

**66 NOTES / 22 TRIADS**

Scale: Bb C D Eb F G A Bb C D Eb F G A Bb

Triads: Bb | Cm | Dm | Eb | F | Gm | Adim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------5--|------6--|
B  |---------|---------|---------|---------|------5--|------7--|---5--8--|---7-----|5--8-----|7--------|8--------|
G  |---------|------5--|------7--|---5--8--|---7-----|5--8-----|7--------|8--------|---------|---------|---------|
D  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
A  |---5--8--|---6-----|5--8-----|6--------|8--------|---------|---------|---------|---------|---------|---------|
E  |6--------|8--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |6--------|5--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---8-----|---6-----|8--------|6--------|---------|---------|---------|---------|---------|---------|---------|
G  |------8--|------7--|---8--5--|---7-----|8--5-----|7--------|5--------|---------|---------|---------|---------|
D  |---------|---------|---------|------8--|------7--|---8--5--|---7-----|8--5-----|7--------|5--------|---------|
A  |---------|---------|---------|---------|---------|---------|------8--|------6--|---8--5--|---6-----|8--5-----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------8--|------6--|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

One triad per beat while learning; two triads per beat as sextuplets after the sequence is secure.

# Level 19 - Major Scale Diatonic Triads: Two Octaves

**Rungs 03-04: play each card as one 22-triad cell, then move +1 fret**

For the descending pass, read the cards in reverse order and move -1 fret

## 03 B MAJOR

**ROOT E7 | FRAME 6-9**

**66 NOTES / 22 TRIADS**

Scale: B C# D# E F# G# A# B C# D# E F# G# A# B

Triads: B | C#m | D#m | E | F# | G#m | A#dim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------6--|------7--|
B  |---------|---------|---------|---------|---------|---------|---------|------7--|------9--|---7-----|---9-----|
G  |---------|---------|---------|---------|------6--|---8-----|---6--9--|---8-----|6--9-----|8--------|9--------|
D  |---------|------6--|------8--|---6--9--|---8-----|6--9-----|8--------|9--------|---------|---------|---------|
A  |---6--9--|---7-----|6--9-----|7--------|9--------|---------|---------|---------|---------|---------|---------|
E  |7--------|9--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |7--------|6--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---9-----|---7-----|9--------|7--------|---------|---------|---------|---------|---------|---------|---------|
G  |------9--|------8--|---9--6--|---8-----|9--6-----|8--------|6--------|---------|---------|---------|---------|
D  |---------|---------|---------|------9--|------8--|---9--6--|---8-----|9--6-----|8--------|6--------|---------|
A  |---------|---------|---------|---------|---------|---------|------9--|------7--|---9--6--|---7-----|9--6-----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------9--|------7--|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

## 04 C MAJOR

**ROOT E8 | FRAME 7-10**

**66 NOTES / 22 TRIADS**

Scale: C D E F G A B C D E F G A B C

Triads: C | Dm | Em | F | G | Am | Bdim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------7--|------8--|
B  |---------|---------|---------|---------|------7--|------9--|---7--10-|---9-----|7--10----|9--------|10-------|
G  |---------|------7--|------9--|---7--10-|---9------|7--10----|9--------|10-------|---------|---------|---------|
D  |---7--10-|---8-----|7--10----|8--------|10-------|---------|---------|---------|---------|---------|---------|
A  |8--------|10-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |8--------|7--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---10----|---8-----|10-------|8--------|---------|---------|---------|---------|---------|---------|---------|
G  |------10-|------9--|---10-7--|---9------|10-7-----|9--------|7--------|---------|---------|---------|---------|
D  |---------|---------|---------|------10-|------9--|---10-7--|---9-----|10-7-----|9--------|7--------|---------|
A  |---------|---------|---------|---------|---------|---------|------10-|------8--|---10-7--|---8-----|10-7-----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------10-|------8--|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

# Level 19 - Major Scale Diatonic Triads: Two Octaves

**Rungs 05-06: play each card as one 22-triad cell, then move +1 fret**

For the descending pass, read the cards in reverse order and move -1 fret

## 05 Db MAJOR

**ROOT E9 | FRAME 8-11**

**66 NOTES / 22 TRIADS**

Scale: Db Eb F Gb Ab Bb C Db Eb F Gb Ab Bb C Db

Triads: Db | Ebm | Fm | Gb | Ab | Bbm | Cdim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------8--|------9--|
B  |---------|---------|---------|---------|---------|---------|---------|------9--|------11-|---9-----|---11----|
G  |---------|---------|---------|---------|------8--|------10-|---8--11-|---10----|8--11----|10-------|11-------|
D  |---------|------8--|------10-|---8--11-|---10----|8--11----|10-------|11-------|---------|---------|---------|
A  |---8--11-|---9-----|8--11----|9--------|11-------|---------|---------|---------|---------|---------|---------|
E  |9--------|11-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |9--------|8--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---11----|---9-----|11-------|9--------|---------|---------|---------|---------|---------|---------|---------|
G  |------11-|------10-|---11-8--|---10----|11-8-----|10-------|8--------|---------|---------|---------|---------|
D  |---------|---------|---------|------11-|------10-|---11-8--|---10----|11-8-----|10-------|8--------|---------|
A  |---------|---------|---------|---------|---------|---------|------11-|------9--|---11-8--|---9-----|11-8-----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------11-|------9--|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

## 06 D MAJOR

**ROOT E10 | FRAME 9-12**

**66 NOTES / 22 TRIADS**

Scale: D E F# G A B C# D E F# G A B C# D

Triads: D | Em | F#m | G | A | Bm | C#dim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------9--|------10-|
B  |---------|---------|---------|---------|------9--|------11-|---9--12-|---11----|9--12----|11-------|12-------|
G  |---------|------9--|------11-|---9--12-|---11----|9--12----|11-------|12-------|---------|---------|---------|
D  |---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
A  |---9--12-|---10----|9--12----|10-------|12-------|---------|---------|---------|---------|---------|---------|
E  |10-------|12-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |10-------|9--------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---12----|---10----|12-------|10-------|---------|---------|---------|---------|---------|---------|---------|
G  |------12-|------11-|---12-9--|---11----|12-9-----|11-------|9--------|---------|---------|---------|---------|
D  |---------|---------|---------|------12-|------11-|---12-9--|---11----|12-9-----|11-------|9--------|---------|
A  |---------|---------|---------|---------|---------|---------|------12-|------10-|---12-9--|---10----|12-9-----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------12-|------10-|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

# Level 19 - Major Scale Diatonic Triads: Two Octaves

## Rungs 07-08: play each card as one 22-triad cell, then move +1 fret

For the descending pass, read the cards in reverse order and move -1 fret

### 07 Eb MAJOR

**ROOT E11 | FRAME 10-13**

**66 NOTES / 22 TRIADS**

Scale: Eb F G Ab Bb C D Eb F G Ab Bb C D Eb

Triads: Eb | Fm | Gm | Ab | Bb | Cm | Ddim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------10-|------11-|
B  |---------|---------|---------|---------|---------|---------|---------|------11-|------13-|---11----|---13----|
G  |---------|---------|---------|---------|-------10-|-------12-|---10-13-|---12----|10-13----|12-------|13-------|
D  |---------|------10-|------12-|---10-13-|---12----|10-13----|12-------|13-------|---------|---------|---------|
A  |---10-13-|---11----|10-13----|11-------|13-------|---------|---------|---------|---------|---------|---------|
E  |11-------|13-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |11-------|10-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---13----|---11----|13-------|11-------|---------|---------|---------|---------|---------|---------|---------|
G  |------13-|------12-|---13-10-|---12----|13-10----|12-------|10-------|---------|---------|---------|---------|
D  |---------|---------|---------|------13-|------12-|---13-10-|---12----|13-10----|12-------|10-------|---------|
A  |---------|---------|---------|---------|---------|---------|------13-|------11-|---13-10-|---11----|13-10----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------13-|------11-|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

### 08 E MAJOR

**ROOT E12 | FRAME 11-14**

**66 NOTES / 22 TRIADS**

Scale: E F# G# A B C# D# E F# G# A B C# D# E

Triads: E | F#m | G#m | A | B | C#m | D#dim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------11-|------12-|
B  |---------|---------|---------|---------|---------|-------13-|---11-14-|---13----|11-14----|13-------|14-------|
G  |---------|---------|---------|-------11-|------13-|---11-14-|---13----|11-14----|13-------|14-------|---------|
D  |---------|------11-|------13-|---11-14-|---13----|11-14----|13-------|14-------|---------|---------|---------|
A  |---11-14-|---12----|11-14----|12-------|14-------|---------|---------|---------|---------|---------|---------|
E  |12-------|14-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |12-------|11-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---14----|---12----|14-------|12-------|---------|---------|---------|---------|---------|---------|---------|
G  |------14-|------13-|---14-11-|---13----|14-11----|13-------|11-------|---------|---------|---------|---------|
D  |---------|---------|---------|------14-|------13-|---14-11-|---13----|14-11----|13-------|11-------|---------|
A  |---------|---------|---------|---------|---------|---------|------14-|------12-|---14-11-|---12----|14-11----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------14-|------12-|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

One triad per beat while learning; two triads per beat as sextuplets after the sequence is secure.

# Level 19 - Major Scale Diatonic Triads: Two Octaves

**Rungs 09-10: play each card as one 22-triad cell, then move +1 fret**

For the descending pass, read the cards in reverse order and move -1 fret

## 09 F MAJOR

**ROOT E13 | FRAME 12-15**

**66 NOTES / 22 TRIADS**

Scale: F G A Bb C D E F G A Bb C D E F

Triads: F | Gm | Am | Bb | C | Dm | Edim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------12-|------13-|
B  |---------|---------|---------|---------|---------|---------|---------|------13-|------15-|---13----|---15----|
G  |---------|---------|---------|---------|---------12-|------14-|---12-15-|---14----|12-15----|14-------|15-------|
D  |---------|------12-|------14-|---12-15-|---14----|12-15----|14-------|15-------|---------|---------|---------|
A  |---12-15-|---13----|12-15----|13-------|15-------|---------|---------|---------|---------|---------|---------|
E  |13-------|15-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |13-------|12-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---15----|---13----|15-------|13-------|---------|---------|---------|---------|---------|---------|---------|
G  |------15-|-------14-|---15-12-|---14----|15-12----|14-------|12-------|---------|---------|---------|---------|
D  |---------|---------|---------|------15-|-------14-|---15-12-|---14----|15-12----|14-------|12-------|---------|
A  |---------|---------|---------|---------|---------|---------|------15-|-------13-|---15-12-|---13----|15-12----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------15-|------13-|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

## 10 Gb MAJOR

**ROOT E14 | FRAME 13-16**

**66 NOTES / 22 TRIADS**

Scale: Gb Ab Bb Cb Db Eb F Gb Ab Bb Cb Db Eb F Gb

Triads: Gb | Abm | Bbm | Cb | Db | Ebm | Fdim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------13-|------14-|
B  |---------|---------|---------|---------|---------|---------|---------|------14-|------16-|---14----|---16----|
G  |---------|---------|---------|---------|------13-|------15-|---13-16-|---15----|13-16----|15-------|16-------|
D  |---------|------13-|------15-|---13-16-|---15----|13-16----|15-------|16-------|---------|---------|---------|
A  |---13-16-|---14----|13-16----|14-------|16-------|---------|---------|---------|---------|---------|---------|
E  |14-------|16-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |14-------|13-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---16----|---14----|16-------|14-------|---------|---------|---------|---------|---------|---------|---------|
G  |------16-|-------15-|---16-13-|---15----|16-13----|15-------|13-------|---------|---------|---------|---------|
D  |---------|---------|---------|------16-|-------15-|---16-13-|---15----|16-13----|15-------|13-------|---------|
A  |---------|---------|---------|---------|---------|---------|------16-|-------14-|---16-13-|---14----|16-13----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------16-|------14-|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

One triad per beat while learning; two triads per beat as sextuplets after the sequence is secure.

# Level 19 - Major Scale Diatonic Triads: Two Octaves

**Rungs 11-12: play each card as one 22-triad cell, then move +1 fret**

For the descending pass, read the cards in reverse order and move -1 fret

## 11 G MAJOR

**ROOT E15 | FRAME 14-17**

**66 NOTES / 22 TRIADS**

Scale: G A B C D E F# G A B C D E F# G

Triads: G | Am | Bm | C | D | Em | F#dim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|------14-|------15-|
B  |---------|---------|---------|---------|---------|---------|---------|------15-|------17-|---15----|---17----|
G  |---------|---------|---------|---------|------14-|------16-|---14-17-|---16----|14-17----|16-------|17-------|
D  |---------|------14-|------16-|---14-17-|---16----|14-17----|16-------|17-------|---------|---------|---------|
A  |---14-17-|---15----|14-17----|15-------|17-------|---------|---------|---------|---------|---------|---------|
E  |15-------|17-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |15-------|14-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---17----|---15----|17-------|15-------|---------|---------|---------|---------|---------|---------|---------|
G  |------17-|------16-|---17-14-|---16----|17-14----|16-------|14-------|---------|---------|---------|---------|
D  |---------|---------|---------|------17-|------16-|---17-14-|---16----|17-14----|16-------|14-------|---------|
A  |---------|---------|---------|---------|---------|---------|------17-|------15-|---17-14-|---15----|17-14----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------17-|------15-|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

## 12 Ab MAJOR

**ROOT E16 | FRAME 15-18**

**66 NOTES / 22 TRIADS**

Scale: Ab Bb C Db Eb F G Ab Bb C Db Eb F G Ab

Triads: Ab | Bbm | Cm | Db | Eb | Fm | Gdim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I--------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------15-|------16-|
B  |---------|---------|---------|---------|--------15-|------17-|---15-18-|------16-|-----18-|---16----|---18----|
G  |---------|---------|---------|---------|------15-|------17-|---15-18-|---17----|15-18----|17-------|18-------|
D  |---------|------15-|------17-|---15-18-|---17----|15-18----|17-------|18-------|---------|---------|---------|
A  |---15-18-|---16----|15-18----|16-------|18-------|---------|---------|---------|---------|---------|---------|
E  |16-------|18-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |16-------|15-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---18----|---16----|18-------|16-------|---------|---------|---------|---------|---------|---------|---------|
G  |------18-|------17-|---18-15-|---17----|18-15----|17-------|15-------|---------|---------|---------|---------|
D  |---------|---------|---------|------18-|------17-|---18-15-|---17----|18-15----|17-------|15-------|---------|
A  |---------|---------|---------|---------|---------|---------|------18-|------16-|---18-15-|---16----|18-15----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------18-|------16-|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

# Level 19 - Final Rung and Full Round-Trip Challenge

## 13 A MAJOR - OCTAVE

**ROOT E17 | FRAME 16-19**

**66 NOTES / 22 TRIADS**

Scale: A B C# D E F# G# A B C# D E F# G# A

Triads: A | Bm | C#m | D | E | F#m | G#dim

**ASCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|I-------|ii-------|iii------|IV-------|V--------|vi-------|vii-dim--|I--------|ii-------|iii------|IV-------|
Deg|1--3--5--|2--4--6--|3--5--7--|4--6--8--|5--7--9--|6--8--10-|7--9--11-|8--10-12-|9--11-13-|10-12-14-|11-13-15-|
Fng|2--1--4--|4--2--1--|1--4--3--|2--1--4--|4--3--1--|1--4--3--|3--1--4--|4--3--2--|1--4--4--|3--2--1--|4--4--2--|
e  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------16-|------17-|
B  |---------|---------|---------|---------|---------|---------|---------|------17-|------19-|---17----|---19----|
G  |---------|---------|---------|---------|------16-|------18-|---16-19-|---18----|16-19----|18-------|19-------|
D  |---------|------16-|------18-|---16-19-|---18----|16-19----|18-------|19-------|---------|---------|---------|
A  |---16-19-|---17----|16-19----|17-------|19-------|---------|---------|---------|---------|---------|---------|
E  |17-------|19-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
```

**DESCENDING - 11 TRIADS / 33 NOTES**

```
Hrm|IV-------|iii------|ii-------|I--------|vii-dim--|vi-------|V--------|IV-------|iii------|ii-------|I--------|
Deg|15-13-11-|14-12-10-|13-11-9--|12-10-8--|11-9--7--|10-8--6--|9--7--5--|8--6--4--|7--5--3--|6--4--2--|5--3--1--|
Fng|2--4--4--|1--2--3--|4--4--1--|2--3--4--|4--1--3--|3--4--1--|1--3--4--|4--1--2--|3--4--1--|1--2--4--|4--1--2--|
e  |17-------|16-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
B  |---19----|---17----|19-------|17-------|---------|---------|---------|---------|---------|---------|---------|
G  |------19-|------18-|---19-16-|---18----|19-16----|18-------|16-------|---------|---------|---------|---------|
D  |---------|---------|---------|------19-|------18-|---19-16-|---18----|19-16----|18-------|16-------|---------|
A  |---------|---------|---------|---------|---------|---------|------19-|------17-|---19-16-|---17----|19-16----|
E  |---------|---------|---------|---------|---------|---------|---------|---------|---------|------19-|------17-|
```

*Return to the low root; release pressure; move the entire frame one fret.*

**Triad pulse: D-U-D / U-D-U; do not reset**

## Full round-trip challenge

Start at A major on low-E fret 5. Play rungs 01-13 without restarting the pick cycle. Reverse after the fret-17 A-major cell and return through rungs 12-01. Say each key name aloud before its first low root. Repeat the complete route with the opposite starting stroke.

| | |
|---|---|
| **ROUTE COUNT** | 25 cells x 66 notes = 1,650 picked notes. |
| **TEMPO RULE** | Add 2 BPM after 3 clean trips; reduce 4-6 BPM after 2 errors. |
| **PICK STARTS** | Complete one D-start round and one U-start round. |
| **HARMONY GOAL** | Name or sing major, minor, and diminished triads without losing the pulse. |

## Clean-round standard

- No missing, doubled, reversed, or reordered triad.
- Every group preserves the 1-3-5 spacing and the one-degree root motion.
- Correct string changes, B-string offset, and assigned fingering.
- Both articulated high-root notes are equal in volume.
- No scrape or tempo change during any one-fret shift.
- Major, minor, and diminished qualities remain audible inside one connected line.

# Level 19 - Practice Architecture, Harmony Training, and Mastery Gate

## Five-phase practice sequence

1. **First-octave checkpoint**
   A major: triads 1-3-5 through 4-6-8, then reverse.
2. **Ascending half**
   A major: all 11 triads / 33 notes. Use both pick starts.
3. **Descending half**
   A major: all 11 triads / 33 notes. Preserve the root motion.
4. **Link the turn**
   Play all 66 notes; articulate both high-root notes without a reset.
5. **Full ladder**
   Use 2-3 neighboring keys, then climb to fret 17 and return.

## Harmony and crossing audit

| | |
|---|---|
| **Triad quality** | Hear major, minor, or diminished; do not reduce the group to a fret pattern. |
| **Root motion** | Each new triad begins one scale degree higher on the up pass and lower on the return. |
| **Internal spacing** | Each chord tone skips one scale degree: root-third-fifth. |
| **String crossing** | Keep pick depth shallow and do not accent the landing string. |
| **Same-finger transfer** | Move repeated finger 4 cleanly between strings rather than smearing two notes. |
| **Frame shift** | Move only after note 66; release pressure before moving one fret. |

## After the basic ladder is clean

1. Accent the root of every triad, then remove all group accents.
2. Accent only the fifth of each triad without rushing into the next root.
3. Say the Roman numeral before each triad in one selected key.
4. Name the actual chords in one selected key: A, Bm, C#m, D, E, F#m, G#dim.
5. Play one triad per beat, then two triads per beat as sextuplets at the same metronome marking.
6. Begin selected sessions at fret 17 and descend first, or choose a random root fret from 5-17.

*These are Level 19 variations, not separate full levels.*

## Mastery gate before Level 20

Complete the full route with a steady triplet pulse, correct triad qualities, silent one-fret shifts, and both pick starts on two separate days.

### Practice log

Date _____ BPM _____ D/U _____ Clean Y/N _____

Date _____ BPM _____ D/U _____ Clean Y/N _____

Date _____ BPM _____ D/U _____ Clean Y/N _____